

A Moore-type bound for local Dictator-to-XOR bijections and a negative answer to a question of Johnston and Scott

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Abstract

Johnston and Scott asked whether there is a bijection ϕ of $\{0, 1\}^n$ mapping Dictator to XOR which is C -Lipschitz, has every output bit depending on at most k input bits, and whose inverse is $o(\log n / \log k)$ -Lipschitz as $n, k \rightarrow \infty$. We show that there is not: every k -local bijection from Dictator to XOR satisfies $n \leq 1 + (k-1) + \dots + (k-1)^{D-1}$, where D is the Lipschitz constant of ϕ^{-1} , with no hypothesis at all on the forward Lipschitz constant; in particular $D \geq 1 + \log n / \log k$. The bound is exact: the linear $(k-1)$ -ary tree maps of Rao and Shinkar attain it for every n and every $k \geq 2$, so the least possible inverse Lipschitz constant is $\lceil \log((k-2)n+1) / \log(k-1) \rceil$ for $k \geq 3$ and n for $k = 2$, and nonlinear maps offer no improvement over linear ones. This sharpens Theorem 1 of Johnston and Scott from $\Omega(1/(k + \log C))$ to $\Omega(1/\log k)$ for general maps.

This note was produced by AI within the [Graph Theory AI](#) project: the argument was found by a language model and audited by an adversarial LLM referee, and it has not been checked by a human mathematician. [Section 5](#) records the full provenance.

1 Introduction

Let $Q_n = \{0, 1\}^n$ with the Hamming distance d_H , let $e_1, \dots, e_n$ be the unit vectors and $\oplus$ the coordinatewise sum modulo 2; for $x \in Q_n$ and $T \subseteq [n]$ write $x^T = x \oplus \bigoplus_{i \in T} e_i$ for x with the coordinates in T flipped. The two Boolean functions of interest are Dictator(x) = x_1 and XOR(x) = $\bigoplus_{i=1}^n x_i$. Following Rao and Shinkar [2] and Johnston and Scott [1], a bijection $\phi: Q_n \rightarrow Q_n$ is a *mapping from f to g* if $f(x) = g(\phi(x))$ for every x . Thus ϕ is a mapping from Dictator to XOR precisely when

$$\bigoplus_{a=1}^n \phi_a(x) = x_1 \quad \text{for every } x \in Q_n. \quad (1)$$

The j -th output bit ϕ_j *depends on* the i -th input bit if $\phi_j(x) \neq \phi_j(x \oplus e_i)$ for some x , and ϕ is k -local if every output bit depends on at most k input bits [1]. Writing $P_a \subseteq [n]$ for the set of input bits on which ϕ_a depends, k -locality says $|P_a| \leq k$ for all a , and $\phi_a(x) = \phi_a(x')$ whenever x and x' agree on P_a (flip the coordinates outside P_a one at a time). A map is C -Lipschitz if $d_H(\phi(x), \phi(x')) \leq C d_H(x, x')$; its Lipschitz constant $\text{Lip}(\phi)$ is the least such C . Since any two points of Q_n are joined by a geodesic of cube edges, $\text{Lip}(\phi) = \max_{x,i} d_H(\phi(x), \phi(x \oplus e_i))$ is the maximum stretch of an edge, hence a nonnegative integer. Throughout, for a bijection ϕ we write

$$D = \text{Lip}(\phi^{-1}),$$

and for integers $k \geq 1$ and $d \geq 1$ we put

$$N_k(d) = \sum_{r=0}^{d-1} (k-1)^r, \quad \text{so that } N_1(d) = 1, \quad N_2(d) = d, \quad N_k(d) = \frac{(k-1)^d - 1}{k-2} \quad (k \geq 3), \quad (2)$$

with the convention $0^0 = 1$. Note that $N_k(d)$ is the number of vertices of the complete rooted $(k-1)$ -ary tree of height $d-1$, so bounds of the form $n \leq N_k(D)$ are Moore-type bounds.

Rao and Shinkar [2] observed that $\phi(x) = (\text{XOR}(x), x_2, \dots, x_n)$ is a 2-bi-Lipschitz mapping from Dictator to XOR but not k -local for any $k < n$, that the 2-local map $(x_1 \oplus x_2, \dots, x_{n-1} \oplus x_n, x_n)$ has an inverse which is not even $(n-1)$ -Lipschitz, and they constructed a linear 3-local 2-Lipschitz mapping from Dictator to XOR with $O(\log n)$ -Lipschitz inverse; their Question 6.1 asked whether a mapping could be both $O(1)$ -local and $O(1)$ -bi-Lipschitz. Johnston and Scott answered this in the negative:

Theorem 1.1 ([1, Theorem 1]). *Let $\phi: \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a mapping from Dictator to XOR which is C -Lipschitz and where each output bit depends on at most k input bits. Then there is a constant $\delta = \Omega(1/(k + \log(C)))$ such that the inverse map ϕ^{-1} is not $\delta \log(n)$ -Lipschitz. Furthermore, if ϕ is a linear mapping, then we may take $\delta = \Omega(1/\log(C + k))$.*

Writing, as in [1], $\delta(C, k)$ for the best constant in Theorem 1.1, Rao and Shinkar's construction (a complete $(k-1)$ -ary tree, [2, Theorem 2 and Appendix A]) shows $\delta(2, k) = O(1/\log k)$, which Theorem 1.1 matches for linear maps; for general maps it only gives $\delta(2, k) = \Omega(1/k)$. At the end of [1, §2] the authors ask, in running prose:

Question 1.2 ([1, §2]). Does there exist a map ϕ from Dictator to XOR which is C -Lipschitz, where each output bit depends on at most k input bits and such that ϕ^{-1} is $o(\log n / \log k)$ -Lipschitz as $n, k \rightarrow \infty$?

The question is restated, still open, in Johnston's thesis [3]; the only citing paper we know of, Boczowski and Shinkar [4], concerns average stretch and does not touch it. We answer Question 1.2 in the negative, by a lower bound that uses neither the forward Lipschitz constant C nor any linearity:

Theorem 1.3 (Main theorem). *Let $n \geq 1, k \geq 1$, and let $\phi: Q_n \rightarrow Q_n$ be a k -local bijection from Dictator to XOR with $D = \text{Lip}(\phi^{-1})$. Then for every $y \in Q_n$ and every integer $d \geq 1$,*

$$\#\{a \in [n] : d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq d\} \leq N_k(d). \quad (3)$$

In particular, taking $d = D$,

$$n \leq N_k(D) = 1 + (k-1) + \dots + (k-1)^{D-1}. \quad (4)$$

The bound is attained, by a construction that is due to Rao and Shinkar [2, Theorem 2 and Appendix A] and is reproduced in Section 3 for completeness.

Theorem 1.4 (Rao–Shinkar tree maps). *Let $k \geq 2, d \geq 1$ and $1 \leq n \leq N_k(d)$. There is a linear k -local bijection $\phi: Q_n \rightarrow Q_n$ from Dictator to XOR with $\text{Lip}(\phi) \leq 2$ and $\text{Lip}(\phi^{-1}) \leq d$.*

Corollary 1.5 (Exact optimum). *Let $k \geq 2$ and $n \geq 1$. The least value of $\text{Lip}(\phi^{-1})$ over all k -local bijections $\phi: Q_n \rightarrow Q_n$ from Dictator to XOR is*

$$D_{\min}(n, k) = \min\{d \geq 1 : n \leq N_k(d)\} = \begin{cases} n, & k = 2, \\ \lceil \log((k-2)n + 1) / \log(k-1) \rceil, & k \geq 3, \end{cases} \quad (5)$$

and it is attained by a linear map with $\text{Lip}(\phi) \leq 2$. The same value is optimal if a forward Lipschitz bound $C \geq 2$ is imposed, while no such bijection is 1-Lipschitz when $n \geq 2$. For $k = 1$ no such bijection exists when $n \geq 2$.

Proof. If ϕ is as in the statement, Theorem 1.3 gives $n \leq N_k(D)$, so $D \geq D_{\min}(n, k)$; Theorem 1.4 with $d = D_{\min}(n, k)$ provides a linear ϕ with $\text{Lip}(\phi) \leq 2$ and $\text{Lip}(\phi^{-1}) \leq D_{\min}(n, k)$, hence equality. By Lemma 3.1 below, imposing $C \geq 2$ excludes nothing and $C = 1$ is impossible for $n \geq 2$. For $k = 2$, $N_2(d) = d$ gives $D_{\min} = n$. For $k \geq 3$, $n \leq N_k(d)$ is equivalent to $(k-2)n + 1 \leq (k-1)^d$, i.e. to $d \geq \log((k-2)n + 1) / \log(k-1)$, and this ratio is at least 1 since $(k-2)n + 1 \geq k-1$. Finally $N_1(D) = 1$ gives $n \leq 1$ when $k = 1$. $\square$

Corollary 1.6 (Answer to [Question 1.2](#)). *Let $k \geq 2$. Every k -local bijection $\phi: Q_n \rightarrow Q_n$ from Dictator to XOR satisfies*

$$\text{Lip}(\phi^{-1}) \geq 1 + \frac{\log n}{\log k}. \quad (6)$$

Consequently the answer to [Question 1.2](#) is no, for every choice of C : along any sequence of such maps with $n, k \rightarrow \infty$ one has $\text{Lip}(\phi^{-1}) > \log n / \log k$, which is not $o(\log n / \log k)$. In the notation of [1], ϕ^{-1} is not $(\log n / \log k)$ -Lipschitz, so $\delta(C, k) \geq 1 / \log k$ for all C and all $k \geq 2$; together with Rao and Shinkar's construction, $\delta(2, k) = \Theta(1 / \log k)$ for general maps, exactly as for linear maps.

Proof. Since $\binom{D-1}{r} \geq 1$ for $0 \leq r \leq D-1$,

$$N_k(D) = \sum_{r=0}^{D-1} (k-1)^r \leq \sum_{r=0}^{D-1} \binom{D-1}{r} (k-1)^r = k^{D-1}.$$

So (4) gives $n \leq k^{D-1}$, i.e. $(D-1) \log k \geq \log n$, which is (6) as $\log k > 0$. The remaining assertions are restatements: $1 + \log n / \log k > \log n / \log k$ for all n, k , so the ratio $\text{Lip}(\phi^{-1}) / (\log n / \log k)$ stays above 1 along every sequence. $\square$

Idea of proof (of [Theorem 1.3](#)). Fix y , let $x = \phi^{-1}(y)$, and for each output index a let $R_a \subseteq [n]$ be the set of input coordinates on which $\phi^{-1}(y)$ and $\phi^{-1}(y \oplus e_a)$ differ, so that $\phi(x^{R_a}) = y \oplus e_a$. The sets R_a are nonempty, pairwise distinct, all contain the coordinate 1 (by (1), flipping any output bit flips x_1), and have size at most D ([Lemma 2.1](#)). The bound is a branching count over the sets R_a . For a nonempty $T \subseteq [n]$ let $A_d(T)$ be the set of a with $T \subseteq R_a$ and $|R_a| \leq d$. Since $x^T \neq x$, some output bit b has $\phi_b(x^T) \neq \phi_b(x)$; then P_b meets T , so $|P_b \setminus T| \leq k-1$. For every $a \in A_d(T)$ other than b , the output ϕ_b takes the same value at x^{R_a} as at x (only bit $a \neq b$ changed), hence a different value than at x^T ; as x^{R_a} and x^T differ exactly on $R_a \setminus T$, the set P_b must meet $R_a \setminus T$, i.e. $a \in A_d(T \cup \{i\})$ for some $i \in P_b \setminus T$. Thus $A_d(T)$ is covered by $\{b\}$ and at most $k-1$ sets $A_d(T \cup \{i\})$, and induction on $d - |T|$ gives $|A_d(T)| \leq N_k(d - |T| + 1)$ ([Lemma 2.2](#)). Taking $T = \{1\}$ gives (3), and $d = D$ gives (4). The single choice of b for each T , made once and not per a , is what yields the branching factor $k-1$ independent of d ; the recursion is a union bound over a covering, so overlaps among the sets $A_d(T \cup \{i\})$ can only help. [\[full proof\]](#)

2 The counting argument

Throughout this section $\phi: Q_n \rightarrow Q_n$ is a k -local bijection from Dictator to XOR, $D = \text{Lip}(\phi^{-1})$, and we fix $x \in Q_n$ and put $y = \phi(x)$. For $a \in [n]$ define

$$x^{(a)} = \phi^{-1}(y \oplus e_a), \quad R_a = \{i \in [n] : x_i \neq x_i^{(a)}\}. \quad (7)$$

Lemma 2.1 (Inverse edges at a point). (i) $x^{(a)} = x^{R_a}$ and $R_a \neq \emptyset$ for every a ;
(ii) the sets $R_1, \dots, R_n$ are pairwise distinct;
(iii) $1 \in R_a$ for every a ;
(iv) $|R_a| = d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq D$ for every a .

Idea of proof. (i) and (iv) are the definitions together with injectivity of ϕ^{-1} . (ii): $R_a = R_b$ forces $x^{(a)} = x^{(b)}$, and applying ϕ gives $a = b$. (iii): by (1) at $x^{(a)}$ and at x , $x_1^{(a)} = \bigoplus_b (y \oplus e_a)_b = x_1 \oplus 1$. [\[full proof\]](#)

For an integer $d \geq 1$ and a nonempty $T \subseteq [n]$ with $|T| \leq d$ let

$$A_d(T) = \{a \in [n] : T \subseteq R_a \text{ and } |R_a| \leq d\}. \quad (8)$$

Lemma 2.2 (Branching bound). *For every $d \geq 1$ and every nonempty $T \subseteq [n]$ with $|T| \leq d$,*

$$|A_d(T)| \leq N_k(d - |T| + 1). \quad (9)$$

Idea of proof. Induction on $d - |T|$. If $|T| = d$ then $R_a = T$ for all $a \in A_d(T)$, so $|A_d(T)| \leq 1$ by Lemma 2.1(ii). If $|T| < d$, pick b with $\phi_b(x^T) \neq \phi_b(x)$; the covering $A_d(T) \subseteq \{b\} \cup \bigcup_{i \in P_b \setminus T} A_d(T \cup \{i\})$ with $|P_b \setminus T| \leq k - 1$ and the induction hypothesis give $1 + (k - 1)N_k(d - |T|) = N_k(d - |T| + 1)$. [\[full proof\]](#)

3 Sharpness: the Rao–Shinkar tree maps

The construction in this section is that of Rao and Shinkar [2, Theorem 2 and Appendix A]: there the tree is a complete binary tree, giving a 3-local map, and they note that a complete $(L - 1)$ -ary tree on n vertices gives an L -local 2-Lipschitz map with C -Lipschitz inverse whenever $(L - 1)^C \geq n$. The only thing added here is bookkeeping: truncating the complete $(k - 1)$ -ary tree in breadth-first order shows that the map exists exactly when $n \leq N_k(d)$, which is what makes the optimum in Corollary 1.5 exact rather than asymptotic.

Idea of proof (of Theorem 1.4). The complete rooted $(k - 1)$ -ary tree of height $d - 1$ has $N_k(d)$ vertices; its first n vertices in breadth-first order form a rooted tree $\mathcal{T}$ on $[n]$ with root 1, at most $k - 1$ children per vertex and height at most $d - 1$. Put $\phi_v(x) = x_v \oplus \bigoplus_{w \text{ child of } v} x_w$. Each ϕ_v reads at most k inputs; in $\bigoplus_v \phi_v$ every non-root input appears twice (in its own and in its parent’s output) and the root once, giving (1); the inverse is $x_v = \bigoplus_{w \in \text{sub}(v)} y_w$, the sum over the subtree of v ; flipping x_v changes at most the outputs v and its parent, so $\text{Lip}(\phi) \leq 2$; flipping y_w changes exactly the inputs at w and its ancestors, so $\text{Lip}(\phi^{-1}) = 1 + \text{height}(\mathcal{T}) \leq d$. [\[full proof\]](#)

Lemma 3.1. *If $n \geq 2$, every bijection $\phi: Q_n \rightarrow Q_n$ from Dictator to XOR has $\text{Lip}(\phi) \geq 2$. In particular the maps of Theorem 1.4 have $\text{Lip}(\phi) = 2$ for $n \geq 2$.*

Idea of proof. For $i \neq 1$, x and $x \oplus e_i$ have the same first coordinate, so by (1) their images have the same parity and lie at even distance; injectivity makes the distance positive. [\[full proof\]](#)

4 Remarks

Remark 4.1 (Interpretation and scope). The definitions used here are those of [1] verbatim: the mapping goes from Dictator to XOR in the sense $\text{Dictator}(x) = \text{XOR}(\phi(x))$, locality is the paper’s global junta condition (each output bit depends on at most k input bits over the whole cube), and Lipschitz constants are with respect to Hamming distance, i.e. maximum edge stretch. The referee (Appendix B) checked these against arXiv:1812.09215v3 and judged the reading fair and intended; the original writeup’s self-declared caveat “full-cube bijections and global k -junta locality” is simply the paper’s setting. Theorem 1.3 is stronger than what Question 1.2 asks for in two ways: no hypothesis on C is used, and the bound is pointwise (3), not only global. It concerns maximum stretch only; it says nothing about average stretch [4], about embeddings into larger cubes, or about maps from XOR to Dictator, none of which is asked.

Remark 4.2 (Why the argument beats Theorem 1.1). Johnston and Scott build one global bipartite dependency graph between input and output bits, bound the degree of an input vertex by $C2^{k-1}$ (a uniformly random witness pays a factor 2^{1-k} for each dependent output), and lose a further factor of two to bipartiteness; this is where $\Omega(1/(k + \log C))$ comes from. The argument of Section 2 instead branches locally: at a partial change T of the input it fixes a single output bit b that has already moved, and every other target must move b back, which forces its R_a to meet P_b outside T , a set of size at most $k - 1$. Neither the forward constant nor the random witness is needed. The recursion (9) is a union bound over a covering, not a count of disjoint levels of a ball; the failure mode of Moore-type

counts that silently assume disjointness therefore does not arise, since overlap only decreases $|A_d(T)|$. The referee re-derived every step, including this point, and could not break it.

Remark 4.3 (Boundary cases). $k = 0$ is vacuous: a 0-local map is constant, hence not injective. For $k = 1$, [Theorem 1.3](#) gives $n \leq N_1(D) = 1$; equivalently, a 1-local bijection is a signed coordinate permutation, whose output XOR is $\bigoplus_i x_i \oplus \text{const} \neq x_1$ for $n \geq 2$. For $k = 2$, (4) reads $n \leq D$, and it is tight for the path map $(x_1 \oplus x_2, \dots, x_{n-1} \oplus x_n, x_n)$ of [2], which is [Theorem 1.4](#) for the path and whose inverse has stretch exactly n (Johnston and Scott note that it “is not even $(n-1)$ -Lipschitz”). For $n = 1$, $\phi = \text{id}$ has $D = 1 = N_k(1)$. For $D \leq 2$, (4) gives $n \leq N_k(2) = k$: a k -local Dictator-to-XOR bijection with 2-Lipschitz inverse needs some output bit reading all n inputs. This last statement, together with the affineness of ϕ^{-1} in that case, was proved directly in the earlier attempt on this problem by gpt-5.6-sol (its Proposition 4; see [Section 5](#)), and is the only part of the present result that attempt reached.

Remark 4.4 (The referee’s computational checks). The scripts are in `verification_astra/scripts/1812.09215__00/` and encode the definitions from scratch. (a) The tree map of [Theorem 1.4](#) was built for all $1 \leq n \leq 14$, $2 \leq k \leq 7$ (78 maps) and checked over the whole cube to be a bijection satisfying (1), k -local, with $\text{Lip}(\phi) = 2$ for $n \geq 2$ and $\text{Lip}(\phi^{-1}) = D_{\min}(n, k)$ exactly in every case, e.g. $(n, k) = (7, 3) \mapsto 3$, $(8, 3) \mapsto 4$, $(13, 4) \mapsto 3$, $(14, 4) \mapsto 4$, $(n, 2) \mapsto n$. (b) Among all $8! = 40320$ permutations of Q_3 there are 576 bijections from Dictator to XOR; those that are 2-local all have $D = 3$ and the 3-local ones have $D \in \{2, 3\}$, with no violation of (4) or of the pointwise bound (3). (c) All bijections from Dictator to XOR with 2-Lipschitz inverse were enumerated for $n = 2, \dots, 6$ (after a harmless normalisation there are $n!$ of them), and the minimum over them of the largest junta size was 2, 3, 4, 5, 6 respectively, i.e. exactly $n = N_k(2)$ with no slack; in particular no 3-local such bijection of Q_4 has a 2-Lipschitz inverse. (d) An exhaustive search over all k -local bijections from Dictator to XOR (nonlinear included) found minimum D equal to $D_{\min}(n, k)$ in every completed case: $(n, k) = (3, 2)$: 48 maps, $D_{\min} = 3$; $(3, 3)$: 576, 2; $(4, 2)$: 1152, 4; $(4, 3)$: 124416, 3; $(5, 2)$: 46080, 5. (e) For the 576 maps at $(3, 3)$ and the 1152 maps at $(4, 2)$, [Lemma 2.1\(ii\),\(iii\)](#) and the full inequality (9) were checked at every base point x and every admissible T : 0 violations, with 5376 resp. 73728 instances of equality in (3). (f) The identities in (2), formula (5) and $N_k(d) \leq k^{d-1}$ were checked numerically for $2 \leq k \leq 59$, $d \leq 39$ and $n \leq 3999$. No check failed. The referee notes that the smallest case where the theorem forbids $D = 3$ at depth, $(n, k) = (8, 3)$, is beyond brute force, so the computations probe the branching factor harder than the depth of the recursion.

Remark 4.5 (Novelty). The referee found no earlier appearance of the lower bound (4), so [Theorem 1.3](#) and [Corollary 1.6](#) appear to be new; the question was open in [1] and in [3]. The construction of [Theorem 1.4](#) is not new: it is Rao and Shinkar’s [2, Theorem 2 and Appendix A], which Johnston and Scott already cite as the source of $\delta(2, k) = O(1/\log k)$, and the original writeup presented it without attribution (corrected here). The novelty of this note is thus the lower bound, and, as a consequence, the exact value (5). [Corollary 1.6](#) strictly implies [Theorem 1.1](#) for general maps, since $1/\log k \geq 1/k \geq 1/(k + \log C)$ for $k \geq 2$, $C \geq 1$. As with every result of this project, and all the more so for a two-page elementary resolution of a published open question, the argument awaits a human referee.

5 Provenance

The mathematics in this note was produced without human intervention, in three stages. (1) The lower bound and its proof were found by the language model gpt-6-astra (reasoning effort maximal) in a single autonomous pass on 2026-09-17, as part of a sweep over open problems catalogued from arXiv (catalog id 1812.09215__00, leg attacks_retry; original writeup in `attacks_retry/1812.09215__00/output.md`). This was a *second* attempt on the problem: the prompt included, as unverified context, an earlier attempt by gpt-5.6-sol (leg attacks, verdict “partial”), which had obtained $D = \Omega(\min\{k, \sqrt{\log n / \log(2Ck)}\})$ by a dependency-graph packing argument with a C -dependent degree bound, together with a complete

description of the case $D \leq 2$ (its Proposition 4: ϕ^{-1} affine and $k \geq n$). The second writeup states that it retains that attempt's inverse-edge set-up, namely the points $x^{(a)} = \phi^{-1}(y \oplus e_a)$, the sets R_a and the observation $1 \in R_a$ (Lemma 2.1(iii); §3.3 of the first attempt), and replaces the packing argument by the branching count of Lemma 2.2; the covering relation, Lemma 2.2, the exact optimum and the answer to Question 1.2 are new to the second attempt. Nothing inherited from the first attempt is presented here as new. (2) An independent adversarial referee agent (claude-opus-5) reviewed the writeup on 2026-09-17: it checked the definitions and the wording of Question 1.2 and Theorem 1.1 against arXiv:1812.09215v3 and against Johnston's thesis, re-derived all 22 steps of the writeup (labelling every one VALID), looked specifically for the failure modes of Moore-type counts (overlapping balls, off-by-one in N_k , the cases $k = 1, 2$ and small n , hidden use of linearity), identified the construction of Section 3 as Rao and Shinkar's, searched for follow-up literature, and ran the exhaustive computations summarised above. Its verdict was CONFIRMED with high confidence. Its report is reproduced verbatim in Appendix B. (3) On 2026-09-17 the writeup was rewritten into the present note by claude-fable-5-1: the text was reorganised, with full proofs in Appendix A, and the following changes were made, all of them taken from the referee report or from the cited sources.

- (a) The writeup asserted without proof that the sets R_a are nonempty and pairwise distinct (the referee's Caveat 2); the two-line proof was added as Lemma 2.1(i),(ii).
- (b) The construction of Section 3 is attributed to Rao and Shinkar [2, Theorem 2 and Appendix A] and is no longer presented as new (the referee's Caveat 1); the sentence describing what differs (the exact threshold $n \leq N_k(d)$) was added.
- (c) The case $k = 1$, treated separately in §4 of the writeup by the same argument, is folded into Theorem 1.3 through the convention $N_1(d) = 1$; the equivalence between “depends on at most k input bits” and the existence of the sets P_a used in the proof is spelled out in the introduction.
- (d) The statement of Theorem 1.1, the discussion of $\delta(C, k)$, and the comparison with the proof of [1] were added from the source paper and the referee report; the writeup cited nothing. The bibliography ([1, 2, 3, 4]) was added.
- (e) The remarks on boundary cases ($k = 0$, $k = 2$ and the path map, $n = 1$, $D \leq 2$ and its relation to the first attempt) and the union-bound remark were taken from the referee report; the observation that the tree maps have $\text{Lip}(\phi) = 2$ exactly for $n \geq 2$ combines two statements of the writeup.

No other mathematical content was changed. **No human mathematician has checked this argument.** We circulate it for human review, not as a claim to a resolution.

A Full proofs

Lemma 2.1 (Inverse edges at a point). (i) $x^{(a)} = x^{R_a}$ and $R_a \neq \emptyset$ for every a ;

(ii) the sets $R_1, \dots, R_n$ are pairwise distinct;

(iii) $1 \in R_a$ for every a ;

(iv) $|R_a| = d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq D$ for every a .

Proof. (i) By definition, $x^{(a)}$ and x differ exactly in the coordinates of R_a , so $x^{(a)} = x^{R_a}$. Since $y \oplus e_a \neq y$ and ϕ^{-1} is injective, $x^{(a)} \neq x$, so $R_a \neq \emptyset$.

(ii) If $R_a = R_b$ then $x^{(a)} = x^{R_a} = x^{R_b} = x^{(b)}$, and applying ϕ gives $y \oplus e_a = y \oplus e_b$, hence $a = b$.

(iii) Apply (1) at the point $x^{(a)}$, whose image is $y \oplus e_a$, and then at x , whose image is y :

$$x_1^{(a)} = \bigoplus_{b=1}^n \phi_b(x^{(a)}) = \bigoplus_{b=1}^n (y \oplus e_a)_b = \left(\bigoplus_{b=1}^n y_b \right) \oplus 1 = x_1 \oplus 1.$$

Hence $x_1^{(a)} \neq x_1$, i.e. $1 \in R_a$.

(iv) $|R_a| = d_H(x, x^{(a)}) = d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq \text{Lip}(\phi^{-1}) d_H(y, y \oplus e_a) = D.$ $\square$

Lemma 2.2 (Branching bound). *For every $d \geq 1$ and every nonempty $T \subseteq [n]$ with $|T| \leq d$,*

$$|A_d(T)| \leq N_k(d - |T| + 1). \quad (9)$$

Proof. We induct on $d - |T| \in \{0, 1, \dots, d - 1\}$.

Base case $|T| = d$. If $a \in A_d(T)$ then $T \subseteq R_a$ and $|R_a| \leq d = |T|$, so $R_a = T$. By Lemma 2.1(ii) at most one index a has $R_a = T$, so $|A_d(T)| \leq 1 = N_k(1)$.

Inductive step $|T| < d$. Since $T \neq \emptyset$ we have $x^T \neq x$, and since ϕ is injective, $\phi(x^T) \neq \phi(x)$. Fix, once and for all for this T , an output coordinate b with

$$\phi_b(x^T) \neq \phi_b(x). \quad (10)$$

The points x^T and x agree outside T ; if $P_b \cap T = \emptyset$ they would agree on P_b and ϕ_b would take the same value at both, contradicting (10). Hence $P_b \cap T \neq \emptyset$ and

$$|P_b \setminus T| \leq |P_b| - 1 \leq k - 1. \quad (11)$$

Now let $a \in A_d(T)$ with $a \neq b$. By Lemma 2.1(i), $\phi(x^{R_a}) = \phi(x^{(a)}) = y \oplus e_a$, which agrees with $y = \phi(x)$ in every coordinate other than a ; in particular, as $a \neq b$,

$$\phi_b(x^{R_a}) = \phi_b(x) \neq \phi_b(x^T). \quad (12)$$

Because $T \subseteq R_a$, the points x^{R_a} and x^T differ exactly in the coordinates of $R_a \setminus T$. If $P_b \cap (R_a \setminus T) = \emptyset$, the two points would agree on P_b and ϕ_b would take the same value at both, contradicting (12). So there is $i \in P_b \cap (R_a \setminus T)$; then $i \in P_b \setminus T$, $T \cup \{i\} \subseteq R_a$, $|T \cup \{i\}| = |T| + 1 \leq d$ and $|R_a| \leq d$, i.e. $a \in A_d(T \cup \{i\})$. We have shown the covering relation

$$A_d(T) \subseteq \{b\} \cup \bigcup_{i \in P_b \setminus T} A_d(T \cup \{i\}), \quad (13)$$

where it is irrelevant whether b itself lies in $A_d(T)$. Each $T \cup \{i\}$ is nonempty with $|T \cup \{i\}| \leq d$ and $d - |T \cup \{i\}| = d - |T| - 1$, so the induction hypothesis gives $|A_d(T \cup \{i\})| \leq N_k(d - |T|)$. Using (13) and (11),

$$\begin{aligned} |A_d(T)| &\leq 1 + \sum_{i \in P_b \setminus T} |A_d(T \cup \{i\})| \\ &\leq 1 + (k - 1)N_k(d - |T|) \\ &= 1 + \sum_{r=0}^{d-|T|-1} (k - 1)^{r+1} = \sum_{r=0}^{d-|T|} (k - 1)^r = N_k(d - |T| + 1). \end{aligned}$$

(For $k = 1$ the union in (13) is empty and the bound is $1 = N_1(d - |T| + 1)$.) This completes the induction. Note that (13) is a covering, not a partition: the sets $A_d(T \cup \{i\})$ may overlap, and the inequality is a union bound. $\square$

Theorem 1.3 (Main theorem). *Let $n \geq 1$, $k \geq 1$, and let $\phi: Q_n \rightarrow Q_n$ be a k -local bijection from Dictator to XOR with $D = \text{Lip}(\phi^{-1})$. Then for every $y \in Q_n$ and every integer $d \geq 1$,*

$$\#\{a \in [n] : d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq d\} \leq N_k(d). \quad (3)$$

In particular, taking $d = D$,

$$n \leq N_k(D) = 1 + (k - 1) + \dots + (k - 1)^{D-1}. \quad (4)$$

Proof. Fix $y \in Q_n$, let $x = \phi^{-1}(y)$ and define R_a by (7), so that $|R_a| = d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a))$ by Lemma 2.1(iv). By Lemma 2.1(iii), $1 \in R_a$ for every a , hence for every $d \geq 1$

$$A_d(\{1\}) = \{a : |R_a| \leq d\} = \{a : d_H(\phi^{-1}(y), \phi^{-1}(y \oplus e_a)) \leq d\}.$$

Lemma 2.2 with $T = \{1\}$ gives $|A_d(\{1\})| \leq N_k(d)$, which is (3). Taking $d = D$, every $a \in [n]$ satisfies $|R_a| \leq D$ by Lemma 2.1(iv), so $A_D(\{1\}) = [n]$ and $n \leq N_k(D)$, which is (4). (Here $D \geq 1$ since $R_a \neq \emptyset$.) $\square$

Theorem 1.4 (Rao–Shinkar tree maps). *Let $k \geq 2$, $d \geq 1$ and $1 \leq n \leq N_k(d)$. There is a linear k -local bijection $\phi: Q_n \rightarrow Q_n$ from Dictator to XOR with $\text{Lip}(\phi) \leq 2$ and $\text{Lip}(\phi^{-1}) \leq d$.*

Proof. The tree. The complete rooted $(k-1)$ -ary tree of height $d-1$ has $(k-1)^r$ vertices at level r for $0 \leq r \leq d-1$, hence $N_k(d)$ vertices in all. List its vertices in breadth-first order (level by level) and keep the first $n \leq N_k(d)$ of them. A vertex's parent lies on a lower level and so precedes it in the list; hence the kept set is closed under taking parents and forms a rooted tree $\mathcal{T}$ on n vertices with root the level-0 vertex, at most $k-1$ children per vertex, and height at most $d-1$. Identify $V(\mathcal{T})$ with $[n]$ so that the root is 1; for $v \in [n]$ let $\text{ch}(v)$ be its set of children and $\text{sub}(v)$ the vertex set of the subtree rooted at v (including v).

The map. Define $\phi: Q_n \rightarrow Q_n$ by

$$\phi_v(x) = x_v \oplus \bigoplus_{w \in \text{ch}(v)} x_w \quad (v \in [n]). \quad (14)$$

This is linear over $\mathbb{F}_2$, and ϕ_v depends only on the $1 + |\text{ch}(v)| \leq 1 + (k-1) = k$ coordinates $\{v\} \cup \text{ch}(v)$, so ϕ is k -local.

Dictator to XOR. Every non-root vertex w has exactly one parent, so in $\bigoplus_v \phi_v(x) = \bigoplus_v x_v \oplus \bigoplus_v \bigoplus_{w \in \text{ch}(v)} x_w$ each x_w with $w \neq 1$ appears exactly twice and x_1 exactly once; thus $\bigoplus_v \phi_v(x) = x_1$, which is (1).

Bijectivity and the inverse. Let $y = \phi(x)$ and fix v . Summing (14) over $w \in \text{sub}(v)$, the children of vertices of $\text{sub}(v)$ are exactly the vertices of $\text{sub}(v) \setminus \{v\}$, each occurring once, so

$$\bigoplus_{w \in \text{sub}(v)} y_w = \bigoplus_{w \in \text{sub}(v)} x_w \oplus \bigoplus_{u \in \text{sub}(v) \setminus \{v\}} x_u = x_v. \quad (15)$$

Thus $\psi(y)_v := \bigoplus_{w \in \text{sub}(v)} y_w$ satisfies $\psi \circ \phi = \text{id}$; so ϕ is an injective self-map of the finite set Q_n , hence a bijection, with $\phi^{-1} = \psi$.

Forward stretch. By (14), flipping x_v changes ϕ_v and, if $v \neq 1$, ϕ_p for the parent p of v , and no other output; so every edge is stretched by at most 2 and $\text{Lip}(\phi) \leq 2$.

Inverse stretch. By (15), flipping y_w changes x_v exactly for those v with $w \in \text{sub}(v)$, i.e. for $v = w$ and the ancestors of w : these are $\text{depth}(w) + 1$ coordinates. Hence $\text{Lip}(\phi^{-1}) = \max_w (\text{depth}(w) + 1) = 1 + \text{height}(\mathcal{T}) \leq d$. $\square$

Lemma 3.1. *If $n \geq 2$, every bijection $\phi: Q_n \rightarrow Q_n$ from Dictator to XOR has $\text{Lip}(\phi) \geq 2$. In particular the maps of Theorem 1.4 have $\text{Lip}(\phi) = 2$ for $n \geq 2$.*

Proof. Let $n \geq 2$, pick $i \in \{2, \dots, n\}$ and any $x \in Q_n$. The points x and $x \oplus e_i$ have the same first coordinate, so by (1) the images $\phi(x)$ and $\phi(x \oplus e_i)$ have the same parity $\bigoplus_a \phi_a$; two points of Q_n of the same parity are at even Hamming distance. As ϕ is injective the distance is positive, hence at least 2, and $\text{Lip}(\phi) \geq 2$. For the maps of Theorem 1.4 this combines with $\text{Lip}(\phi) \leq 2$. $\square$

B Report of the LLM referee (verbatim)

The following is the unedited report of the adversarial referee agent (claude-opus-5, 2026-09-17) on the *original* writeup. Its step numbers and equation numbers refer to that writeup, not to this note. The scripts it mentions are in `verification_astra/scripts/1812.09215__00/`. Treat it as machine output, not as an authority.

B.1 Interpretation

The writeup’s formalisation is exactly the one in the source paper. I checked the definitions against the arXiv v3 text of Johnston–Scott (arXiv:1812.09215):

- *Mapping from f to g* (paper, §1): “we say that a bijection $\varphi : \{0,1\}^n \rightarrow \{0,1\}^n$ is a mapping from f to g if, for every $x \in \{0,1\}^n$, we have $f(x) = g(\varphi(x))$.” With $f = \text{Dictator}$, $g = \text{XOR}$ this is precisely the writeup’s (1): $\oplus_a \varphi_a(x) = x_1$. The direction is the correct one (not the reverse).
- *k-local* (paper, §1): “the j th output bit φ_j depends on the i th input bit x_i if there exists $x \in \{0,1\}^n$ such that $\varphi_j(x) \neq \varphi_j(x+e_i)$. If every output bit depends on at most k input bits, we say the map φ is k -local.” This is the writeup’s $P_a \subseteq [n]$, $|P_a| \leq k$, and it is a *global* junta condition in the paper too. The writeup’s self-declared caveat (“uses full-cube bijections and global k -junta locality”) is therefore not a caveat at all: it is the paper’s own setting.
- *Lipschitz* is w.r.t. Hamming distance, and on the cube the Lipschitz constant of a map equals its maximum edge stretch (concatenate a geodesic), so $D = \text{Lip}(\varphi^{\{-1\}})$ is a positive integer. Correct.
- The question, verbatim from the paper (and restated verbatim in Johnston’s Oxford thesis, ORA uuid:970b5eb2, p. ~2260 of the text layer): “Does there exist a map φ from Dictator to XOR which is C -Lipschitz, where each output bit depends on at most k input bits and such that $\varphi^{\{-1\}}$ is $o(\log n / \log k)$ -Lipschitz as $n, k \rightarrow \infty$?”

The writeup proves $D \geq 1 + \log n / \log k$ for **every** such φ , with no hypothesis on C at all. That is strictly stronger than what is needed to answer the question, and it answers it in the negative in the sense a reasonable author would accept. There is no interpretation gaming: no hypothesis is strengthened, no strawman reading is refuted, and the result is not sensitive to any convention (height/depth, indexing, the choice of the dictator coordinate). **Interpretation: fair and intended.**

One genuine scope remark: the writeup answers the *maximum-stretch* question. It says nothing about average stretch, and nothing about maps into a larger cube. Neither is asked.

B.2 Step-by-step findings

Numbering follows the writeup. $N_k(d) := \sum_{r=0}^{d-1} (k-1)^r$.

#	step	label	note
1	§1, (1): formalisation of “Dictator $\rightarrow$ XOR bijection”	VALID	matches the paper’s definition verbatim (see Interpretation).
2	§1: “the Lipschitz constant equals the maximum stretch of an edge, so D is an integer”	VALID	standard geodesic/telescoping argument on the cube; needed later only to say “
3	§2: $R_a := \{i : x_i \neq x^{\{a\}}_i\}$ are nonempty and pairwise distinct	VALID (stated without proof, one line)	the n points $\varphi^{\{-1\}}(y \oplus e_a)$ are distinct and all $\neq x$ since $\varphi^{\{-1\}}$ is injective; R_a determines $x^{\{a\}}$ given x . Trivially repairable, so I do not count it as a gap.
4	§2, (5): $1 \in R_a$ for every a	VALID	$x^{\{a\}}_1 = \oplus_b (y \oplus e_a)_b = (\oplus_b y_b) \oplus 1 = x_1 \oplus 1$ by (1). Re-derived.
5	Claim (6), base case $ T = d$	VALID	$a \in A_d(T)$ forces $T \subseteq R_a$ and
6	Claim (6), inductive step: choice of b with $\varphi_b(x^{\wedge}T) \neq \varphi_b(x)$	VALID	$T \neq \emptyset \implies x^{\wedge}T \neq x \implies \varphi(x^{\wedge}T) \neq \varphi(x)$ by injectivity, so such a b exists.
7	Claim (6), (8): $P_b \cap T \neq \emptyset$ hence $ P_b \cap T \leq k-1$	VALID	$x^{\wedge}T$ and x agree off T ; if $P_b \cap T = \emptyset$ then φ_b would agree, contradicting step 6. Then
8	Claim (6), (9): for $a \in A_d(T)$, $a \neq b$, $\varphi_b(x^{\wedge}\{R_a\}) = \varphi_b(x) \neq \varphi_b(x^{\wedge}T)$	VALID	$\varphi(x^{\wedge}\{R_a\}) = \varphi(x^{\wedge}\{a\}) = y \oplus e_a$, which agrees with y in every coordinate $\neq a$.
9	Claim (6): $x^{\wedge}\{R_a\}$ and $x^{\wedge}T$ differ exactly on $R_a \cap T$, hence $P_b \cap (R_a \cap T) \neq \emptyset$	VALID	$T \subseteq R_a$ gives $R_a \Delta T = R_a \cap T$; if P_b missed it, φ_b would agree with itself at the two points, contradicting step 8. (En passant this shows $R_a \neq T$ for $a \neq b$, consistent with step 5.)
10	Claim (6), covering relation (10)	VALID	every $a \in A_d(T) \setminus \{b\}$ lies in $A_d(T \cup \{i\})$ for some $i \in P_b \cap T$, with
11	Claim (6), the arithmetic $1 + (k-1)N_k(d- T) = N_k(d- T +1)$	VALID	$1 + \sum_{r=0}^{d-1} (k-1)^r$

#	step	label	note
12	§2: this is a union bound, not an injective tree count	VALID — and this is the point the referee brief flagged	the worry “the counted ball may overlap itself” does not apply: (10) is a <i>covering</i> of $A_d(T)$ by $\leq 1 + (k-1)$ sets, so overlap can only make $A_d(\{1\}) = \{a :$
13	§2, (11) and $n \leq N_k(D)$	VALID	$A_d(\{1\}) = \{a :$
14	§3: existence of the tree (BFS truncation of the complete $(k-1)$ -ary tree of height $d-1$) whenever $n \leq N_k(d)$	VALID	$N_k(d)$ is exactly the number of vertices of that complete tree; truncating in BFS order keeps $\leq k-1$ children and height $\leq d-1$. Verified in code for $n \leq 14, k \leq 7$.
15	§3, (12)–(13): locality ($\leq k$), Dictator $\rightarrow$ XOR, bijectivity, $\text{Lip}(\varphi) = 2$, $\text{Lip}(\varphi^{\wedge\{1\}}) = \text{maxdepth} + 1 \leq d$	VALID (but prior art , see Reference check)	all five properties re-verified by brute force over the whole cube for $n \leq 14, k \leq 7$ — no failures.
16	§3: $\text{Lip}(\varphi) \geq 2$ for $n \geq 2$	VALID	flipping $x_i, i \neq 1$, preserves x_1 hence preserves output parity, so the output distance is even; injectivity makes it positive, hence ≥ 2 .
17	§4: closed form $n \leq ((k-1)^D - 1)/(k-2)$ for $k \geq 3$; $D_{\min}$ formula (3)	VALID	$n \leq N_k(d) \iff (k-2)n+1 \leq (k-1)^d \iff d \geq \log((k-2)n+1)/\log(k-1)$. Verified against the exact integer $d_{\min}$ for all $2 \leq k \leq 39, 1 \leq n \leq 3999$.
18	§4: $N_k(D) \leq \sum_r C(D-1, r)(k-1)^r = k^{D-1}$, hence $D \geq 1 + \log n/\log k$	VALID	$C(D-1, r) \geq 1$ for $0 \leq r \leq D-1$. Verified numerically for all $2 \leq k \leq 59, 1 \leq D \leq 39$.
19	§4: this excludes $D = o(\log n/\log k)$	VALID	$D \geq 1 + \log n/\log k > \log n/\log k$ for all n, k , so no sequence with $n, k \rightarrow \infty$ can have $D = o(\log n/\log k)$.
20	§4: degenerate case $k = 1$	VALID	(10) with $T = \{1\}$ has $P_b T = \emptyset$, so $n \leq 1$. (Equivalently: a 1-local bijection is a signed coordinate permutation, whose output XOR is $\oplus_i x_i + \text{const} \neq x_1$ for $n \geq 2$.)
21	§1, (3): “this remains the exact optimum if any fixed forward Lipschitz bound $C \geq 2$ is imposed”	VALID	the lower bound is C -free and the construction has $\text{Lip}(\varphi) = 2$; $C = 1$ is impossible for $n \geq 2$ by step 16.
22	§1: “the forward Lipschitz assumption is unnecessary for the lower bound”	VALID	C is indeed never used in §2. This is exactly why the argument beats Johnston–Scott’s Theorem 1, whose left-degree bound $C \cdot 2^{\wedge\{k-1\}}$ is the source of their $1/k$.

No step is labelled GAP or ERROR. I looked specifically for the failure modes the brief named and for the ones this problem invites:

- *Self-overlapping ball count* — not applicable (step 12).
- *Off-by-one in N_k* — checked three ways: by hand, against the tree vertex count, and against exhaustive minima (see below). $N_k(D)$ counts vertices of a $(k-1)$ -ary tree of height $D-1$, and $D = \text{depth} + 1$; the two “+1”s are consistent.
- *$k = 1, 2$ and small n* — step 20, and $k = 2$ gives $N_2(D) = D$, i.e. $n \leq D$, which is exactly tight for Rao–Shinkar’s 2-local path map $\varphi = (x_1 + x_2, \dots, x_{n-1} + x_n, x_n)$ whose inverse the paper says “is not even $(n-1)$ -Lipschitz”.
- *Silent use of linearity* — §2 uses none. The counting is purely combinatorial and applies verbatim to nonlinear maps; my exhaustive searches below include all nonlinear maps in range.

B.3 Reference check

The writeup cites **nothing** — it invokes no external theorem, so there is no misquoted citation to find. I nevertheless checked everything it is implicitly positioned against, by downloading the sources.

1. **arXiv:1812.09215v3 (Johnston–Scott), definitions.** Confirmed verbatim, quoted in the Interpretation section above. The writeup’s (1) and its locality hypothesis are exactly the paper’s.
2. **arXiv:1812.09215v3, Theorem 1.** Confirmed verbatim: “Let $\varphi : \{0,1\}^n \rightarrow \{0,1\}^n$ be a mapping from Dictator to XOR which is C -Lipschitz and where each output bit depends on at most k input bits. Then there is a constant $\delta = \Omega(1/(k + \log(C)))$ such that the inverse map $\varphi^{\wedge\{1\}}$ is not $\delta \log(n)$ -Lipschitz. Furthermore, if φ is a linear mapping, then we may take $\delta = \Omega(1/(\log(C+k)))$.” Their proof (§2) builds one *global* dependency graph, bounds the left-degree by $C \cdot 2^{\wedge\{k-1\}}$ via a uniformly random point (paying $2^{\wedge\{1-k\}}$ for each witness), and loses a further factor 2 to bipartiteness in their Lemma 6. The writeup’s localized branching argument pays neither cost, which is a coherent explanation of why it improves $\Omega(1/(k+\log C))$ to the sharp $\Omega(1/\log k)$. The new bound does **not** contradict Theorem 1; it strictly implies it.
3. **The open question.** Confirmed verbatim in arXiv:1812.09215v3 §2 and, independently, in Tom Johnston’s Oxford DPhil thesis “Problems in Extremal and Probabilistic Combinatorics” (ORA uuid:970b5eb2), where it is restated word for word with no progress reported. So the problem really was open.
4. **arXiv:1501.03016 (Rao–Shinkar), Theorem 2 and Appendix A — PRIOR ART FOR §3.** This is the one substantive finding of the reference check. Rao–Shinkar’s Appendix A constructs $A = I + M$ where M is the adjacency matrix of a complete **binary tree**

with the root labelled 1, proves $A^{\{-1\}}$ is the ancestor/descendant matrix, and notes the resulting map is 3-local, 2-Lipschitz with $O(\log n)$ -Lipschitz inverse. They then write, verbatim: “Note that the above proof can be generalized to obtain a mapping φ that is L -local 2-Lipschitz such that $\varphi^{\{-1\}}$ is C -Lipschitz, for any L and C that satisfy $(L-1)^C \geq n$ by replacing the tree G in the proof with a complete $L-1$ -ary tree on n vertices.” That is *exactly* the writeup’s §3 (its (12) is $A = I + M$ and its (13) is $A^{\{-1\}}$ = ancestor matrix), and Johnston–Scott cite it as the source of $\delta(2,k) = O(1/\log k)$. The writeup presents §3 as its own with no attribution. The only thing new in §3 is the BFS truncation, which turns Rao–Shinkar’s $(k-1)^d \geq n$ into the exact threshold $n \leq N_k(d)$ — a genuine but very small sharpening, and it is what makes the claimed *exact* optimum (3) exact rather than merely asymptotic.

5. **Follow-up literature.** Semantic Scholar lists exactly one paper citing arXiv:1812.09215 (“On Mappings on the Hypercube with Small Average Stretch”, Boczkowski–Shinkar, 2019), which concerns average stretch and does not touch this gap. Four web searches turned up no paper resolving $\Omega(1/k)$ vs $O(1/\log k)$. I found no prior appearance of the lower bound $n \leq N_k(D)$. So the **lower bound (§2) is not ALREADY KNOWN**, while the **construction (§3) is**.

B.4 Computational check

Scripts in `verification_astra/scripts/1812.09215__00/`, run with system `python3`. All encode the definitions from scratch (no reuse of the writeup’s formulas except as the prediction being tested). “Dictator→XOR bijection” is checked as $\text{parity}(\varphi(x)) = x_1$ for all x ; junta size is recomputed from the truth table; $D = \max$ over edges of the inverse.

(a) `tree_construction.py` — **the finite object in §3**. Built the BFS-truncated $(k-1)$ -ary tree map for all $1 \leq n \leq 14$, $2 \leq k \leq 7$ (78 maps) and checked, by exhaustive enumeration of the whole cube, that each is a bijection, satisfies $\oplus_v \varphi_v(x) = x_1$, has max junta size $\leq k$, $\text{Lip}(\varphi) = 2$ ($n \geq 2$), and $\text{Lip}(\varphi^{\{-1\}}) \leq d_{\min}$. **Result: 0 failures**, and in every single case $\text{Lip}(\varphi^{\{-1\}})$ equalled $d_{\min}$ exactly — e.g. $(n,k) = (7,3) \rightarrow 3$, $(8,3) \rightarrow 4$, $(13,4) \rightarrow 3$, $(14,4) \rightarrow 4$, $(n,2) \rightarrow n$. The construction is real and exactly attains the bound.

(b) `exhaustive_n3.py` — **every bijection of Q_3** . Enumerated all $8! = 40320$ permutations of Q_3 ; 576 of them are Dictator→XOR bijections. Results: $k = 2 \implies D \in \{3\}$ (formula: 3); $k = 3 \implies D \in \{2,3\}$, min 2 (formula: 2). Violations of $n \leq N_k(D)$: 0. Violations of the pointwise bound (11): 0.

(c) `search_D2.py` — **every map with $D \leq 2$, for $n = 2..6$** . Reparametrising $\psi = \varphi^{\{-1\}}$ as $\psi(y) = (\text{XOR}(y), g(y))$ turns “ $\text{Lip}(\psi) \leq 2$ ” into “ g is 1-Lipschitz and injective on each parity class”, which lets me enumerate *all* such maps (linear and nonlinear) exhaustively. After the harmless normalisation $g(0) = 0$ the counts are $n!$ (2, 6, 24, 120, 720 for $n = 2..6$). For every one of them I recomputed the junta sizes of φ . Minimum over all of max junta size: **2, 3, 4, 5, 6 for $n = 2, 3, 4, 5, 6$** — i.e. exactly n , matching the theorem’s $n \leq N_k(2) = k$ with no slack. In particular there is **no** Dictator→XOR bijection of Q_4 with 3-local outputs and 2-Lipschitz inverse, and none of Q_5 with 4-local outputs and 2-Lipschitz inverse. This is the sharpest available falsification test of the $D = 2$ layer and the theorem passes it.

(d) `search_junta_minD.py` — **every k -local map, small (n,k)** . DFS over the output coordinates with exact fiber-size pruning, last coordinate forced by the Dictator→XOR condition. Completed cases (min D found / formula $d_{\min}$):

n	k	# maps	min D	formula $d_{\min}$
3	2	48	3	3
3	3	576	2	2
4	2	1152	4	4
4	3	124416	3	3
5	2	46080	5	5

Every case matches exactly. The $(n,k) = (4,3)$ row is the most informative: $N_3(2) = 3 < 4$, so the theorem forbids $D = 2$ there, and an exhaustive sweep of all 124416 3-local Dictator→XOR bijections of Q_4 (nonlinear included) finds minimum $D = 3$. The runs for $(6,2)$ and $(4,4)$ were still going when I stopped; they test nothing the completed rows do not.

(e) `pointwise_check.py` — **the Claim itself, not just its corollary**. For every map found in (d) at $(n,k) = (3,3)$ and $(4,2)$ I recomputed the sets R_a at every one of the 2^n base points, asserted $1 \in R_a$ and that the R_a are distinct (steps 3–4 above; **no assertion ever fired**), and checked the full Claim (6) $|A_d(T)| \leq N_k(d - |T| + 1)$ for every nonempty $T \subseteq [n]$ with $|T| \leq d$ and every d , as well as the corollary (11). Results: 576 maps / 0 violations of (11) / 0 violations of (6) / 5376 tight instances of (11); and 1152 maps / 0 / 0 / 73728 tight instances. The $(4,3)$ sweep was still running at cutoff. The large number of *tight* instances shows the bound is not slack-padded.

(f) `formula_check.py` — verified $N_k(d) = ((k-1)^d - 1)/(k-2)$ for $k \geq 3$ and $= d$ for $k = 2$; that $\min\{d : n \leq N_k(d)\}$ equals the boxed formula (3) (both as exact integer arithmetic and as the literal float expression $\lceil \log((k-2)n+1)/\log(k-1) \rceil$, to catch rounding traps); and $N_k(d) \leq k^{\{d-1\}}$. This script was still finishing under heavy machine load at cutoff; the same identities were also re-derived by hand and are confirmed indirectly by the exact agreement of (a)–(d) with the formula across ~90 parameter pairs.

No computational check failed. I specifically hunted for a counterexample at every parameter pair where the theorem is sharp and the search is feasible, and found none.

B.5 Caveats

1. **Unattributed prior art (the main defect).** §3 is Rao–Shinkar’s construction (arXiv:1501.03016, Theorem 2 + the generalisation note in Appendix A), presented as if new. The writeup’s own caveat field says “literature priority has not been independently checked”, which is honest but understates it: the construction half of the “exact optimum” is 2015 work that the source paper explicitly cites. This does not affect correctness, and the *new* half — the lower bound — is, as far as I can determine, not in the literature.
2. **Step 3 asserted, not proved.** “The sets R_a are nonempty and pairwise distinct” is one line of missing argument. Trivial, but it is load-bearing for the base case.
3. **The claimed exactness of (3) is slightly stronger than the question needs** and depends on the BFS-truncation refinement of the construction; the asymptotic answer to the question does not depend on it.
4. **$k = 0$ is not discussed** (vacuous: constant outputs are not injective). **$n = 1$ is fine** ($\varphi = \text{id}$, $D = 1 = N_k(1)$). **$k = 1$ is handled correctly.**
5. **Maximum stretch only.** The result says nothing about average stretch, about embeddings into larger cubes, or about maps from XOR to Dictator. None is asked, but the one-line verdict (“No”) should be read as answering only the maximum-stretch question.
6. **Locality is the paper’s global notion.** If someone instead meant “locally k -junta” (per-region dependence), the argument would need redoing. The paper does not mean that, so this is not a live objection.

7. **The one-line verdict slightly oversells §3:** “linear 2-Lipschitz tree maps attain the resulting bound” is true but is Rao–Shinkar’s, not the writeup’s, contribution.
8. **Extraordinary-claim risk.** This is a two-page elementary argument that settles a question posed in a CPC paper and restated in an Oxford thesis, superseding that paper’s Theorem 1 in the general case. I verified every inference and could not break it, and the exhaustive searches agree with it to the last unit at every sharp threshold I could reach ($n \leq 6$ for $D = 2$, $n \leq 5$ for $k = 2$, $n = 4$ for $k = 3$). But the sharp threshold that would most stress the argument’s *depth* recursion — $(n, k) = (8, 3)$, where the theorem forbids $D = 3$ — is out of brute-force reach, so my computational evidence probes the branching factor much harder than the depth.

B.6 Referee summary

I set out to break this and failed. The load-bearing object is the Claim in §2, and its inductive step survives pedantic re-derivation: fixing a nonempty T with $T \subseteq R_a$ for all a in the family, one picks a *single* output coordinate b whose value already moved between x and $x^{\wedge}T$; every other target a must then move φ_b back, which forces R_a to meet P_b outside T , and $|P_b \setminus T| \leq k-1$ because P_b must already meet T . That is a covering of $A_d(T)$ by $1 + (k-1)$ smaller families, so the recursion $|A_d(T)| \leq 1 + (k-1)|A_d(T \cup \{i\})|$ is a union bound. The specific failure mode the brief asked me to look for — a Moore-type count that silently assumes the levels of a ball are disjoint — does not arise, precisely because this is a covering argument in which overlap only helps. The forward constant C is genuinely never used, which is exactly the source of the improvement over Johnston–Scott’s Theorem 1: their global dependency graph must bound the left-degree by $C \cdot 2^{\{k-1\}}$ (a uniform-measure witness costing $2^{\{1-k\}}$), and they lose another factor 2 to bipartiteness; the localized argument pays neither, so $\Omega(1/(k+\log C))$ becomes the sharp $\Omega(1/\log k)$. Computationally the bound is not merely true but exactly attained everywhere I could check: the tree construction hits $\text{Lip}(\varphi^{\wedge}\{-1\}) = d_{\min}$ for all 78 pairs ($n \leq 14$, $k \leq 7$); an exhaustive sweep of *all* 576 Dictator $\rightarrow$ XOR bijections of Q_3 , all 124416 3-local ones of Q_4 , all 46080 2-local ones of Q_5 , and all maps with 2-Lipschitz inverse for $n \leq 6$ reproduces $\min D = d_{\min}$ in every single case, with zero violations of the pointwise Claim over ~ 1700 maps $\times$ all base points $\times$ all subsets T . The two real deficiencies are scholarly rather than mathematical: §3 reproduces Rao–Shinkar’s $(k-1)$ -ary tree construction without attribution (arXiv:1501.03016, Appendix A, which Johnston–Scott cite as the source of $\delta(2, k) = O(1/\log k)$), so only the lower bound §2 is new, and the assertion that the R_a are distinct is left to the reader. Subject to the usual caveat that an elementary two-page resolution of a published open problem deserves a second human referee — and noting that the deepest sharp case, $(n, k) = (8, 3)$, is beyond brute force — I confirm the writeup: the answer to Johnston–Scott’s question is **no**, and $\delta(2, k) = \Theta(1/\log k)$ for general maps, not just linear ones.

References

- [1] T. Johnston and A. Scott, Lipschitz bijections between boolean functions, *Combin. Probab. Comput.* 30 (2021), 513–525; [arXiv:1812.09215](#).
- [2] S. Rao and I. Shinkar, On Lipschitz bijections between boolean functions, *Combin. Probab. Comput.* 27 (2018), 411–426; [arXiv:1501.03016](#).
- [3] T. Johnston, *Problems in Extremal and Probabilistic Combinatorics: Cubes, Squares and Permutations*, DPhil thesis, University of Oxford, 2021; [ORA uuid:970b5eb2](#).
- [4] L. Boczkowski and I. Shinkar, On mappings on the hypercube with small average stretch, *Combin. Probab. Comput.* (2023); [arXiv:1905.11350](#).